

Course Overview

Database System Concepts

University of Guilan

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Fall 2025

Course Info:

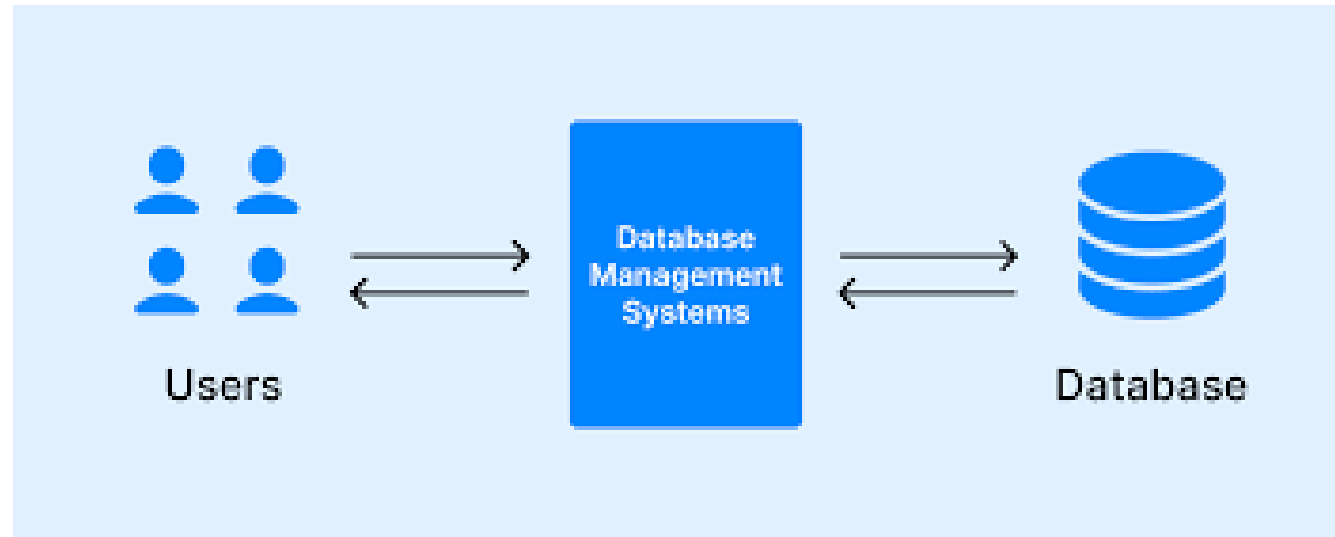
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- Quera:
 - https://quera.org/course/add_to_course/course/23886/
 - Pass:DB-Fall404
- Telegram:
 - <https://t.me/+KwVzzkGoFAg0MTg0>

Why Database Systems?

- Traditional File System:
 - Store data in separate files
 - For each query, scan the entire file
 - Adding new fields is difficult
 - Not scalable for large systems

Database Management System (DBMS)

- Centralized management of data
- Provides controlled access, integrity, recovery
- Hides low-level details → users focus on what, not how



Database Design Process

- Requirements Analysis
- Conceptual Design (ER Modeling)
- Logical Design (Relational Schema)
- Schema Refinement (Normalization)
- Physical Design & Implementation
- Database Usage & Maintenance

Structured Query Language (SQL)

- Standard language for relational databasesUsed to:
 - Define schema (DDL: Data Definition Language)
 - Manipulate data (DML: Data Manipulation Language)
 - Query and retrieve information
 - Control access and security (DCL: Data Control Language)
- Supported by all major DBMS (SQL Server, Oracle, MySQL, PostgreSQL)

Theoretical Foundations

- Database Architecture
 - Levels of abstraction (physical, logical, view)
- Relational Model
 - Relations, keys, constraints
- Schema Refinement
 - Normalization (1NF, 2NF, 3NF, BCNF)
- Integrity Constraints
 - Domain, key, referential, application-specific
- Relational Algebra & Relational Calculus
 - Formal foundations for queries
- Advanced SQL (selected topics)

Assessment

- Midterm Exam: 6 points
 - Final Exam: 7 points
 - Homework / Assignments: 3 points
 - Project: up to 5.5 points (includes 1.5 bonus)
 - Attendance & Participation: +1 bonus
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- Final grade will be calculated out of 20 points, with additional opportunities for bonus

Course Objectives

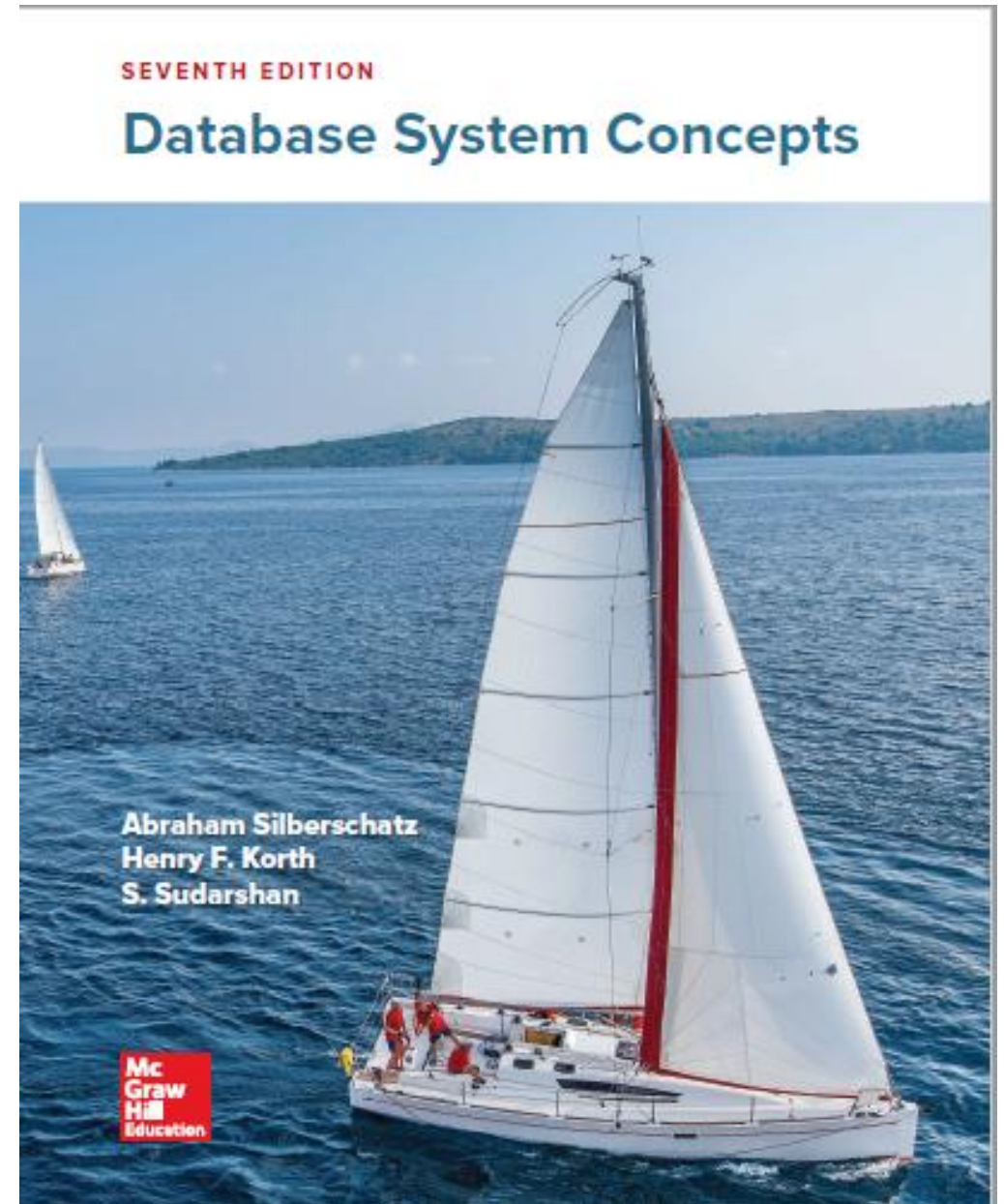
- Build a solid foundation in database systems
- Learn how to model, design, and implement databases
- Acquire practical skills in SQL and query processing
- Understand key principles of data integrity and transactions
- Apply knowledge in a real-world project

Course Outline

- Introduction and Basic Concepts
- Entity-Relationship (ER) Modeling
- Relational Design of ER Models
- SQL: Structured Query Language
- Data Integrity and Security
- Schema Refinement & Normalization

Text Book:

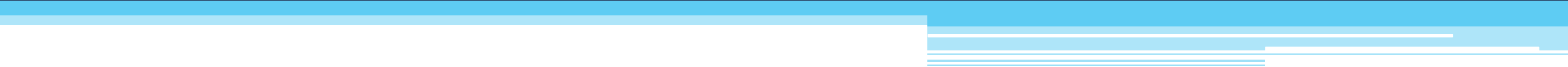
- *Database System Concepts*,
 - *7th Edition*
- Authors: Abraham Silberschatz, Henry F. Korth, and S. Sudarshan
- Publisher: McGraw-Hill, 2019



Other References:

- *Fundamentals of Database Systems* – Elmasri & Navathe
- *Database Management Systems* – Ramakrishnan & Gehrke
- *An Introduction to Database Systems* – C.J. Date
- *Database Lecture Notes* – Dr. Rouhani Rankouhi (**Persian**)
- *Principles of Database Design* – Maghsoumi (**Persian**)

End of Overview

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